Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	27 January 2026
Team ID	LTVIP2026TMIDS79347
Project Name	Online payment fraud detection using machine learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Registration**	Registration through Email & Password Registration through Gmail Registration through LinkedIn
FR-2	**User Confirmation**	Email verification link OTP verification
FR-3	**User Login**	Secure login using email & password Multi-factor authentication (OTP)
FR-4	**Transaction Monitoring**	View transaction history View transaction details Display fraud risk score
FR-5	**Fraud Detection**	ML model predicts transaction as Fraud / Legitimate Real-time fraud scoring

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system should have a simple, user-friendly interface accessible to both technical and non-technical users.
NFR-2	**Security**	All user data must be encrypted (SSL/TLS). Passwords must be hashed. System must prevent unauthorized access.
NFR-3	**Reliability**	The fraud detection system should provide consistent and accurate predictions with minimal failure.
NFR-4	**Performance**	Fraud prediction should be generated within 2–3 seconds for real-time transactions.
NFR-5	**Availability**	The system should have at least 99% uptime to support continuous online transactions.
NFR-6	**Scalability**	The system should handle increasing number of transactions without performance degradation.